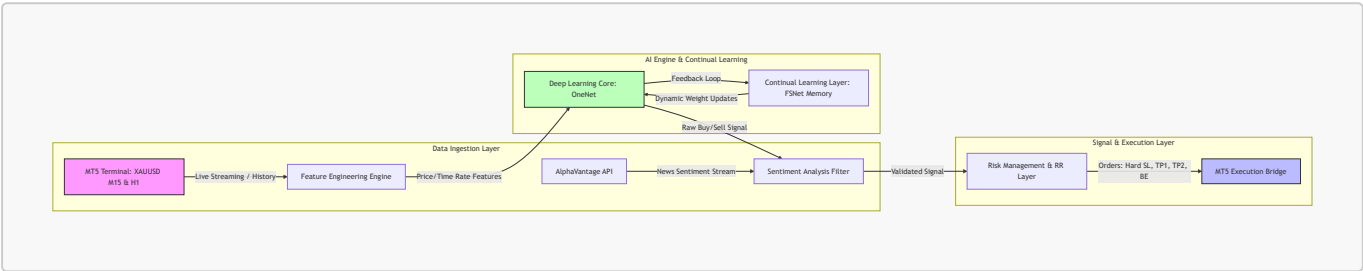


AI Engine for XAU/USD Trading

1. Executive Summary

Automated ML trading engine for XAU/USD using Wave corrections (M15) confirmed by H1 SuperTrend. OneNet/FSNet deep learning with continual learning, AlphaVantage sentiment filter, and MT5 execution. Targets: Monthly Return > 5%, MDD < 10%, validated over 5-year backtest.

2. System Architecture



3. Core Logic & AI Pipeline

3.1 Multi-Timeframe Strategy

- **Macro Filter (H1)**: SuperTrend determines structural direction ($Price\ Rate = \Delta P / \Delta t$, $Time\ Rate = \Delta t$).
- **Execution Trigger (M15)**: Tracks "Fast Motive Bars". Positions are initiated strictly upon the validated completion of an Elliott Wave corrective structure (e.g., Wave 2 or Wave 4 pullbacks) to capture impulsive waves.

3.2 Machine Learning & Sentiment Loop

- **Architecture: OneNet Framework** for stream processing paired with **FSNet (Fast/Slow Networks)** to allow continuous learning from streaming data without catastrophic forgetting.
- **Sentiment Filter**: Real-time NLP processing via **AlphaVantage**. Signals conflicting with the dominant sentiment (Positive/Negative/Neutral) are discarded.

4. Execution & Advanced Risk Management (RR-Based)

To achieve the targeted asymmetric risk profile, position management relies strictly on predefined Risk-to-Reward (RR) multiples rather than fixed pip metrics:

1. **Position Inception**: Trade executed via MT5 API. A server-side **Hard Stop Loss (SL)** is placed at the recent Elliott Wave structural invalidation point ($0.0\ R$). Total risk per trade is capped at 1% of equity.
2. **Take Profit 1 (1.5:1 RR)**: When price reaches a **1.5:1 Risk-to-Reward ratio**, 50% of the volume is closed. The remaining position's SL is automatically moved to **Break-Even (BE)** plus commissions.
3. **Take Profit 2 (3:1 RR) & Trailing**: When price reaches a **3:1 Risk-to-Reward ratio**, an additional 25% of the volume is scaled out. The final 25% activates a **Trailing Stop** tied to the M15 structural swings to capture extended macro moves.

5. System Target Performance Metrics

The complete infrastructure will be benchmarked and rejected if it fails to meet the following parameters during validation:

- **Backtest Horizon:** 5 Years (2021 – 2026) historical walk-forward backtest.
 - **Target Monthly Return:** $> 5\%$ net profit.
 - **Risk Tolerance:** Maximum Peak-to-Trough Drawdown (**MDD**) $< 10\%$.
 - **Data Quality:** 99% Tick Precision data including simulated slippage and dynamic spreads.
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6. Project Budget & Phased Milestones

The total budget for this implementation is **\$1,000 USD** mapped out across two target milestones to verify profitability before going into production:

Milestone 1: Quantitative Strategy Validation & Backtest Report

- **Timeline:** 2 Weeks
- **Budget Allocation:** \$600 USD
- **Deliverables:** * Fully built feature extraction engine (H1 SuperTrend + M15 EW Core Logic).
- Integration of the OneNet and FSNet continual learning frameworks.
- Definitive **5-Year Backtest Report** confirming whether the architecture meets the $> 5\%$ monthly return and $< 10\%$ MDD targets using high-precision tick data.

Milestone 2: MT5 Bridge Integration & Guided Paper Trading

- **Timeline:** 3 Weeks
 - **Budget Allocation:** \$400 USD
 - **Deliverables:**
 - Integration of the AlphaVantage News Sentiment API gate filter.
 - Execution layer coding for the MetaTrader 5 (MT5) python bridge.
 - Implementation of the automated Risk-to-Reward rules (TP1/TP2, Break-Even modifications, Trailing Stops).
 - **3-Week Active Paper Trading Phase:** Multi-week live forward-testing on an MT5 demo account. Collaborative monitoring to catch anomalies, analyze forward drawdowns, and fine-tune model parameters dynamically based on actual market live metrics.
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7. Key Deliverables at Project Completion

1. **Backtest Logs:** A fully reproducible backtesting pipeline validating the 5-year track record.
2. **ML Engine Source Code:** Production-ready Python modules (**OneNet** + **FSNet** framework).
3. **Live MT5 Connector:** Operational execution engine managing the live trade matrix seamlessly on demo/forward environments.